UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 17

Content.

1. Empirical Risk Minimization
2. Classfication with 1D Thresholds

Recall: $S = \{(X_i, Y_i) : 1 \leq i \leq N\}$ is a dataset consisting of iid samples $(X_i, Y_i) \sim P_X \otimes P_{Y|X}$.

1 Empirical Risk Minimization

$$\inf_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^{N} \ell(f(X_i), Y_i)$$

Intuitively, this seeks an f that *fits* the data. Namely, find a function that 'explains the data' as accurately as possible. Here, ℓ is a loss function, where

$$f : \mathcal{X} \to \mathcal{Y} \quad \text{and} \quad \ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}.$$

Let

$$\hat{f} = \arg \inf_{f \in \mathcal{F}} \hat{\mathcal{L}}(f).$$

Quantify the accuracy of 'learnt' $\hat{f}$ on the fresh data.

$$\mathcal{L}(f) = \mathbb{E}_{(X,Y)}[\ell(f(X), Y)]$$

- $\mathcal{L}(\hat{f})$ is a random variable - as $\hat{f}$ is random itself, where the source of randomness is the dataset S.

- For any **fixed** $f^* \in \mathcal{F}$,

$$\mathbb{E}_S[\hat{\mathcal{L}}(f^*)] = \mathbb{E}_S[\frac{1}{N}\sum_{i=1}^{N} \ell(f^*(X), Y)] = \mathbb{E}[\ell(f(X), Y)] = \mathcal{L}(f^*)$$

In fact, $\hat{\mathcal{L}}(f^*) \overset{\text{a.s.}}{\to} \mathcal{L}(f^*)$ by the SLLN. This is why empirical risk is a good proxy for the population loss.
Issue: $\hat{f}$ is a function of data $(X_i, Y_i), 1 \leq i \leq N$. Thus, in general

$$\mathbb{E}_S[\hat{\mathcal{L}}(\hat{f})] \neq \mathbb{E}[L(\hat{f})].$$

Lemma 1. For any function class $\mathcal{F} = \{f : \mathcal{X} \to \mathcal{Y}\}$,

$$\mathbb{E}_S[\mathcal{L}(\hat{f})] - \inf_{f\in\mathcal{F}} \mathcal{L}(f) \leq \mathbb{E}_S[\sup_{f\in\mathcal{F}} L(f) - \hat{\mathcal{L}}(f)]$$

The quantity

$$\mathbb{E}_S\mathcal{L}(\hat{f}) - \inf_{f\in F} \mathcal{L}(f)$$

is known as the *excess risk* associated to empirical risk minimizer $\hat{f}$.

Proof. Let $f^* = \arg\inf_{f\in\mathcal{F}} \mathcal{L}(f)$. In particular f^* is deterministic. We write:

$$\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) = \mathcal{L}(\hat{f}) - \hat{\mathcal{L}}(\hat{f}) + \hat{\mathcal{L}}(\hat{f}) - \hat{\mathcal{L}}(f^*) + \hat{\mathcal{L}}(f^*) - L(f^*)$$

Here $\hat{\mathcal{L}}(\hat{f}) - \hat{\mathcal{L}}(f^*) \leq 0$ since $\hat{f}$ is the minimizer of $\hat{\mathcal{L}}$. So:

$$\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) \leq \mathcal{L}(\hat{f}) - \hat{\mathcal{L}}(\hat{f}) + \hat{\mathcal{L}}(f^*) - \mathcal{L}(f^*).$$

We then arrive at

$$\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) \leq \sup_{f\in\mathcal{F}}(\mathcal{L}(f) - \hat{\mathcal{L}}(f)) + \hat{\mathcal{L}}(f^*) - \mathcal{L}(f^*)$$

Take the expectation with respect to S. Since $\mathbb{E}_S[\hat{\mathcal{L}}(f^*)]\mathcal{L}(f^*)$, we are left with

$$\mathbb{E}_S[L(\hat{f}) - L(f^*)] \leq \mathbb{E}_S[\sup_{f\in\mathcal{F}}(\mathcal{L}(f) - \hat{\mathcal{L}}(f))].$$

This completes the proof of lemma. □

We "sup-out" the dependence of $\hat{f}$ on data. In next several lectures, we will learn various techniques for bounding

$$\mathbb{E}_S\left[\sup_{f\in F}\big(\mathcal{L}(f) - \hat{\mathcal{L}}(f)\big)\right].$$

2 Classification with 1D Thresholds

Loss $\ell(a,b) = \mathbb{1}_{a \neq b}, \quad a = Y, b = f(X)$.
Note the following *trick*: $\mathbb{1}_{a \neq b} = a + (1-2a)b$. With this,

$$\mathbb{E}[\sup_{f \in \mathcal{F}} L(f) - L(f^*)] = \mathbb{E}[\sup_{f \in \mathcal{F}} \mathbb{E}[Y + (1-2Y)f(X)] - \frac{1}{N}\sum_{i=1}^{N} Y_i + (1-2Y_i)f(X_i)]$$

$$= \mathbb{E}[\sup_{f \in \mathcal{F}} \mathbb{E}[(1-2Y)f(X)] - \frac{1}{N}\sum_{i=1}^{N}(1-2Y_i)f(X_i)].$$

Let's ignore the random signs $(1-2Y), (1-2Y_i) \in \{-1, 1\}$ for a while and consider

$$\mathbb{E}[\sup_{f \in \mathcal{F}} \mathbb{E}[f(X)] - \frac{1}{N} f(X_i)]. \quad (*)$$

Example 1. *1D-Thresholds*

$$\mathcal{F} = \{x \to \mathbb{1}_{x \leq \theta} : \theta \in \mathbb{R}\}$$

Let's calculate $(*)$ for this choice.

$$\mathbb{E}[\sup_{\theta \in \mathbb{R}} \mathbb{P}[X \leq \theta] - \sum_{i=1}^{N} \mathbb{1}_{X_i \leq \theta}]$$

Thus, we've arrived at $\mathbb{E}[\sup_{\theta \in \mathbb{R}} F(\theta) - F_N(\theta)]$, which resembles Kolmogorov-Smirnov statistics. Let $U_\theta = F(\theta) - F_N(\theta)$.
Q: How do we control $\mathbb{E}[\sup_{\theta \in \mathbb{R}} U_\theta]$?
Recall: $\sup_{\theta \in \mathcal{B}_2} \langle \theta, x \rangle$. We controlled this using covering number.
Technique 1 *(Bracketing)*

- We 'cover' the range of the distribution
- Assume F has no atoms (i.e the CDF is continuous)
- U_θ is nearly constant on regions of small mass.

Quantiles: a 'cover' $\theta_0, \theta_1, \ldots, \theta_{n+1}$ where we reduce the problem to controlling $\max_{j=0,1,\ldots,n} U_{\theta_j}$.
Introduce $\theta_0 = -\infty$, $\theta_{n+1} = \infty$ and $\mathbb{P}[\theta_i \leq x \leq \theta_{i+1}] = \frac{1}{n+1}$ for $i = 0, \ldots, N$. Fix $\theta \in \mathbb{R}$. Define $\ell(\theta), u(\theta)$ such that

- $\ell(\theta), u(\theta) \in \{\theta_0, \ldots, \theta_{n+1}\}$

- $\ell(\theta) \leq \theta \leq u(\theta)$.

$$U_\theta = \mathbb{P}[X \leq \theta] - \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}_{X_i \leq \theta} \leq P[X \leq u(\theta)] - \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}_{X_i \leq \ell(\theta)}$$

$$\mathbb{P}[\ell(\theta) \leq X \leq u(\theta)] \leq \frac{1}{n+1} \Rightarrow \mathbb{P}[X \leq u(\theta)] \leq \mathbb{P}[X \leq \ell(\theta)] + \frac{1}{n+1}$$

$$U_\theta \leq P[X \leq \ell(\theta)] - \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}_{X_i \leq \ell(\theta)} + \frac{1}{n+1} = U_{\ell(\theta)} + \frac{1}{n+1}$$

$$\sup_{\theta \in \mathbb{R}} U_\theta \leq \max_{j=0,\ldots,n} U_{\theta_j} + \frac{1}{n+1}$$

$$\mathbb{E}[\sup_{\theta \in \mathbb{R}} U_\theta] \leq \mathbb{E}[\max_{j=0,\ldots,n} U_{\theta_j}] + \frac{1}{n+1}$$

$$U_{\theta_j} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{P}[X \leq \theta_j] - \mathbb{1}_{X_i \leq \theta_j}$$

We observe that U_{θ_j} is sub-Gaussian with variance proxy $1/N$. Thus, from earlier lectures,

$$\mathbb{E}[\max_{j=0,\ldots,n} U_{\theta_j}] \leq \frac{1}{\sqrt{N}}\sqrt{2\log(n+1)}.$$

When we choose $n = N - 1$, which is optimal, we obtain

$$\mathbb{E}[\max_{j=0,\ldots,n} U_{\theta_j}] \leq O(\sqrt{\frac{\log N}{N}}).$$

Lemma 2. Let $\mathcal{F} = \{f : \mathcal{X} \to \mathcal{Y}\}$. Let $X_1, \ldots, X_n$ be iid RVs with values in X and let $\mathcal{E}_1, \ldots, \mathcal{E}_n$ be iid Rademacher RVs. Then,

1. $\mathbb{E}[\sup_{f \in \mathcal{F}} \mathbb{E}[f(X)] - \frac{1}{n}\sum_{i=1}^{n} f(x_i)] \leq 2\mathbb{E}[\sup_{f \in \mathcal{F}} \frac{1}{n}\mathcal{E}_i f(X_i)]$
2. $\mathbb{E}[\sup_{f \in \mathcal{F}} \frac{1}{n}\sum_{i=1}^{n} f(x_i) - \mathbb{E}[f(X)]] \leq 2\mathbb{E}[\sup_{f \in \mathcal{F}} \frac{1}{n}\mathcal{E}_i f(X_i)]$
3. $\mathbb{E}[\sup_{f \in \mathcal{F}} |\frac{1}{n}\sum_{i=1}^{n} f(x_i) - \mathbb{E}[f(X)]|] \leq 2\mathbb{E}[\sup_{f \in \mathcal{F}} |\frac{1}{n}\mathcal{E}_i f(X_i)|]$